

CONFIRMATION

THEORIES are pretty easy to come by. Even with the demand for genuinely explanatory theories, there are lots of competing claims about the unseen objects and events that genuinely explain the phenomena we see. But at most one of those theories is true. The urgent responsibility of science, then, is to weed out the bad from the good, that is, to expose the failures in false theories and to champion only those theories that have endured a process of testing that rewards the true and scorns the false. This means that, as much as is possible, a theory must be tested against the observable data. For all a theory's theoretical elegance and pragmatic virtue, there is no reasonable claim to accurate representation of the world until it confronts the world. You've got to do the experiments.

This requirement of empirical testing is probably one's first and most intuitive answer to the question of what makes science believable, and it is a good answer. If the theories claim to be about the world, then they should be accountable to information from the world. Confirmation by testing against observations is the most straightforward external event in science, and achieving a positive test is the most obvious of external virtues. But it is important to be clear on what is meant by "testing" a theory and by "comparison with the observable evidence." Theories usually describe objects and events that are not amenable to observation. That's what makes them interesting, but it is also what blocks us from simply and directly comparing what the theories say to what is observed in the world. Take, for example, the definitive theoretical claim that oxygen atoms have eight protons. Checking

this is not a simple matter of looking and counting; rather, it is a matter of checking on the consequences of the theoretical claim by determining what effect the eight protons must have on the observable world. No one sees the protons or the oxygen atoms, but it might be easy to see some of the effects they have. There is, then, this kind of indirectness in any case of using observational evidence to test a theory. It is a complicating factor worth keeping an eye on when deciding what it means to confirm a theory and whether there is any reason to think that certainty is a reasonable expectation of science. How close can we get to certainty, if not all the way?

To say that explanatory theories are pretty easy to come by is to say that the activity of discovery of theories is not the place to apply strict standards of scientific responsibility. There are lots of ways that people can and have come up with theories to explain phenomena, by observation, sometimes meticulous, sometimes haphazard, or by thinking, comparing and combining other beliefs to see what they suggest, or even by accident as in a dream or hallucination. However a theory is discovered or created, the place to apply rigorous standards is in the subsequent activity of justification. This is where science gets scientific, and it is this context of justification that is the topic here. You can dream up your theory in any way you like, but it must be accountable to observational justification if it is to be taken seriously.

This means not only that a theory must be testable, the internal feature discussed in Chapter 2, but that it must also be tested. Acquiring and evaluating the internal feature of testability is, as a rule, an inexpensive process, requiring only a clear understanding of the theory and the time and energy to do some calculations or whatever reasoning is appropriate for drawing out the consequences and for understanding the causal efficacies of the theory. The theory that describes how the sun works – for example, that the source of energy is an ongoing process of nuclear fusion at the center – is clearly testable. The nuclear fusion must produce huge

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numbers of neutrinos, a species of electrically neutral elementary particles, and some of these neutrinos ought to find their way to the earth where they can be detected. If nuclear fusion is the energy source, the sun must be showering us not only with heat and light but with neutrinos as well. This straightforward and inexpensive reasoning proves the theory to be testable since the neutrinos are detectable. Determining whether or not the neutrinos are in fact detected is another matter. This is the external affair of promoting a testable theory to the status of being a tested theory. Establishing this external virtue is the expensive part of science, for now we have to build the equipment and run the experiments. Consulting the world seems always to cost more than consulting our own thoughts.

This process of testing is known optimistically as confirmation, and in a sense there is no rigid dichotomy between the activities of confirmation and explanation. As will become clear in the following sections, they follow the same logical form, which compares the consequences of theories to observations of the world and thus connects theories with the world by observing the effects that the theorized entities have on objects we can see. If the neutrinos are detected we will say that the neutrinos contribute to the confirmation of the solar theory and that the solar theory explains the shower of neutrinos. It is typical, though, to apply the concept of explanation to cover past observations of events, things we knew happened but wondered why, and reserve the concept of testing to cover future observations, predictions whose outcome is yet unknown. Explanation and confirmation are also distinguished by the pragmatics of the situation, namely, what the scientists happen to be interested in. If it is the phenomenon itself that is puzzling and the question is "Why does this happen?" then it is explanation they are after. If the interest is centered on the status of the theory itself and the phenomena are just hoops it must jump, then it is confirmation. There is, of course, nothing wrong with a combined notion of explanation-as-

confirmation, and this is exactly what we did in Chapter 3 by considering explanation as a truth-conducive virtue. The focus in this chapter, though, will be on the external virtue a theory displays by having true, observable consequences, whether or not they are explanatory. The concern is not so much *why* the phenomena happen as it is *that* they happen as predicted by the theory.

A BASIC MODEL OF CONFIRMATION

Start with an example, Einstein's general theory of relativity. In its early years, as a proposed rival to Newton's well-tested and phenomenally successful account of gravity, the general theory of relativity is appropriately thought of as a hypothesis, a provocative suggestion in want of testing. It is a testable theory insofar as it makes definitive predictions of observable phenomena. It describes gravity as so ubiquitous a force in the universe as to be in the fabric of space and time themselves. Anything moving through space and time is affected by gravity. All material objects, planets, bricks, people, will feel the pull of gravity and will move accordingly, but this is old news; Newton already said as much. The new idea is that even the passage of light, since it moves through space and time, will be affected by gravity. A consequence of the new theory is that light will be deflected if it passes near a massive object since the light will be attracted by the object's gravity. For example, light from a distant star will be very slightly bent if it passes near the sun. The light is bent if it passes near a planet or even a basketball, but of these three only the sun is massive enough to bend the starlight by an amount large enough to detect. With the sun providing the gravity, the bending of light is observable by noting the relative positions of stars when the sun is well out of the way, that is, when there is none of the putative distortion by gravitational bending, and compare this with the relative positions of the same stars when

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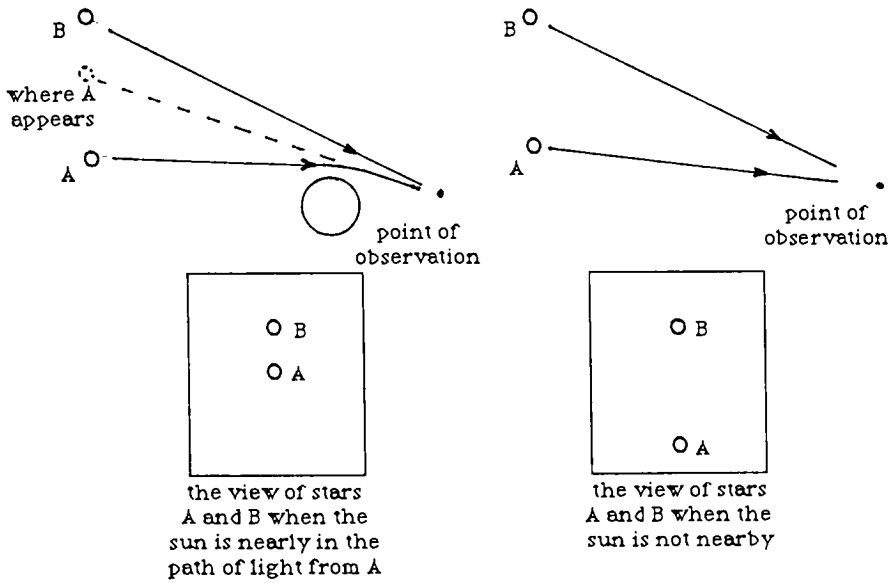


Figure 4.1. Two stars, A and B, viewed in two circumstances to see if the gravity of the sun affects the trajectory of light.

their light rays must pass near the sun (see Figure 4.1). Einstein's theory predicts that the sun, or any sufficiently massive object nearly in the path of light, will cause a distortion in the observed pattern of stars. The effect will be observable only during an eclipse of the sun, not only because no one wants to look right at the sun but also because the sunlight must be blocked for the nearby stars to be visible.

The first total eclipse of the sun to occur after Einstein introduced his theory, the first, that is, to occur during clear weather and without the distraction of a world war, was in 1919. The prediction about the observed positions of the stars was true. This was clearly good news for the theory, and it helped make Einstein everyone's favorite scientist. But just how good is the news? The prediction was confirmed, but was the theory? What's the difference? What does one true prediction prove? Suppose

the prediction turned out to be wrong? Would that have been sufficient evidence to disqualify the theory once and finally?

Clues to answering these questions lie in the basic logical relation between the theory and test results. The initial model we are about to propose will need some fixing up to accommodate realistic examples, but it is a useful place to begin. The theory being tested, Einstein's theory in this example, is regarded in this case as the hypothesis, thus marking it as the claim in question, the one proposed with, as yet, no strong commitment to its being true. Testing it requires that observational consequences be deduced from the hypothesis, reasoning, as in the relativity case, that if the hypothesis is true then the world must behave in such-and-such a way and we should be able to observe this. This draws out the causal consequences of the theory and describes phenomena as determined by the theorized objects and events.

This scheme of finding the deductive consequences of the hypothesis in question is called the **hypothetico-deductive method of confirmation**, H-D for short. The deductive part is only the inference from the hypothesis to its observable consequence. If the hypothesis is true, then the effect must happen. That is, on the assumption that the hypothesis is true, what phenomena must follow? Observing the effect then, seeing that the predicted phenomenon does in fact happen, enhances the credibility of the theory, but it does not constitute unequivocal confirmation. It is not a deductive demonstration that the theory is true. It's not even close. To claim that it is would be to argue in the form that "if the theory (the hypothesis) is true then the effect must happen," and "the effect does, in fact, happen" so "the theory *must* be true." This is exactly the same fallacious form of reasoning encountered in analyzing the truth-conducive properties of explanation and the same fallacious argument used to reason that if it's summertime it must be the fourth of July.

The point is that observing a true consequence of a theory is not deductive, absolutely certain confirmation of the theory.

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Hypothetico-deductive is not intended to mean a deductive proof, nor does it suggest the deducing of the theory from observational evidence. It is rather the other way around, the deducing of observational evidence, predictions, from the theory. True predictions then provide only inductive support for the theory by increasing the likelihood that it is true. If one successful H-D test helps a little, then further H-D tests will help more. The general theory of relativity does not rest its credibility on the single success of predicting the bending of starlight. It makes other predictions, for example, that gravity affects the passage of time and the frequency of light. This is the gravitational red shift, and it too has been observed. In clear hypothetical-deductive form it contributes inductive support to the theory.

This pattern of H-D confirmation is likely to be just an articulation of common sense. Everyone knows that one successful prediction does not prove beyond all doubt that a theory is true. But what about one failed prediction? Does that unequivocally disprove a theory? Suppose that observations during the 1919 eclipse had been contrary to Einstein's prediction.

Since we are still talking about a hypothesis and its deductive predictions, it makes sense, in the case where the prediction fails, to speak of hypothetico-deductive falsification. If the hypothesis is true the effect *must* occur, but the effect does not occur, so the hypothesis must be false. This is a valid deductive argument, and it does prove with certainty that the hypothesis is false. The idea is that if the theorized object or event is *always* supposed to cause the observable effect mentioned, but in fact the effect is missing, then the theorized object must not be there. There is no leeway in this argument, no degrees of unlikelihood. The case is closed. On this model of H-D falsification, a single failed prediction will suffice to falsify a theory and force its rejection.

There is, apparently, an asymmetry in this model of testing between confirmation and falsification. Confirmation comes in degrees and is vulnerable to confusion as to how much is enough.

Falsification, on the other hand, shows no such sloppiness, no degrees, no need for discretion. Once is enough. By virtue of the logical pattern of drawing predictions from theories, falsification has the advantage of a stricter standard, a deductive demonstration of failure. This advantage has motivated some who have come this way before, most notably Karl Popper, to emphasize falsification, that is, weeding out rather than cultivating, as the most crucial phase of the scientific process, and falsifiability as the essential feature of a scientific theory.

This basic model of H-D confirmation and falsification, though, is much too simple. It misses crucial details of real cases of scientific testing, and it is only because of this mistaken view of science that the asymmetry between confirming and falsifying appears. In real life, one failed test does not force the rejection of a theory, and the model of testing must be fixed to show how this works and to see if it is reasonable to retain a theory in the face of failure.

In its favor, though, the simple H-D model does capture the core of common sense about scientific testing. Examples like the general theory of relativity show that H-D gets at least the basics right. It reflects the reasoning of scientists in their plans for experiments to test theories, and many real examples of scientific testing clearly show the essentials of the H-D model. The relatively new suggestion, for example, that the electromagnetic force and the weak nuclear force, the cause of some forms of radioactivity, are in fact two manifestations of a single, electro-weak force needs to be tested. This new theory makes the novel prediction that protons decay. Any proton is likely to remain intact for a very long time, but it could eventually split spontaneously into other particles. The theory precisely describes the decay products and thus suggests observable consequences. To test the theory, look for the characteristic traces of proton decay. Thus from theory to experiment. It is a complicated experiment, requiring keeping tabs on lots of protons, since each one is

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very unlikely to decay. With so many protons and so few decays, the results are, so far, unclear. Experimenters are indeed following the H-D pattern of checking the consequences to test the theory, but observations of the consequences have not been conclusive.

Another good example of hypothetico-deductive testing is the search for solar neutrinos. The standard and well-entrenched theory of the anatomy of stars, including our sun, has it that nuclear fusion is the source of stellar and solar energy. Nuclear fusion is well understood, at least to the point of knowing that the process produces neutrinos which, according to theory, should easily escape from the solar interior where they are born and some of them, a predictable number, should find their way to earth. Furthermore, enough is known about neutrinos to know how to detect them. It isn't easy, but it's theoretically possible. It is done with a huge tank of cleaning fluid, deep underground to provide shielding from other distracting particles. Neutrinos occasionally interact with the chlorine in the cleaning fluid, changing it into a radioactive argon atom. The radioactive emissions from the argon can then be counted as the footprints of solar neutrinos.

This is fine H-D form. If the hypothesis about solar fusion is true, there must be detectable solar neutrinos. So test the theory by looking for the neutrinos. The results in this case are not at all ambiguous. The experiments to detect solar neutrinos have been on the air for years, and yet they have not detected any neutrinos, or at least not nearly as many as the theory predicts. This is a frustrating result that is well known to astronomers and physicists as the solar-neutrino problem, but apparently it is not a big enough problem to force the rejection of the theory, and contrary to the naive model of falsification, the failure to observe the predicted effect has not been taken as conclusive proof that the theory is false. Furthermore, this is not an unusual case of testing in science. There is always something to blame, perhaps in the experimental setup or in the reasoning process that issued the

prediction. It is not, after all, a simple, two-component deduction from the hypothesis about solar fusion to the prediction of clicks on the detector counting radioactive argon. It is a more complicated argument than the simple H-D model indicates, and a realistic model of scientific testing must take the details seriously.

MORE DETAILED AND MORE REALISTIC

To judge by the complication and expense of scientific experiments, the predictions being checked are not simple statements of the form “ x happens.” The prediction usually specifies how to set up the experiment and how to detect the effect as caused by the theorized events. It specifies the proper conditions under which the testing and viewing are to be done. The prediction, in other words, is a conditional statement. *If* things are set up in this way, and *if* the observing conditions are right, *then* we will see x . Recall the balloon theory, the one that explains why rubbed balloons stick to walls by the scraping off of old, crusty rubber, exposing fresh, sticky rubber. To test this theory one might use the theory itself to predict that a rubbed balloon will crust over again if left to dry and so the balloon will no longer stick after a sufficient drying time. The prediction is a conditional, an if–then statement. If the balloon is left alone for a while, then it will no longer stick to the wall. Both the specified condition, leaving the balloon alone, and the result, its not sticking, are events that can be observed.

As with the solar-neutrino experiment, the prediction being checked is complicated. In terms of the objects and events amenable to observation, it says something like this: If you put a tank of cleaning fluid underground, and if you periodically flush it and separate out the argon, and if you count the radioactive decays, then there will be x number of counts. There are a lot of condi-

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tions that must be properly satisfied in order to bring about the final result of meaningful clicks on the counter.

If the results do not happen as expected, if there are too few clicks or if the balloon still sticks to the wall, we *can* blame the conditions. Maybe the experiment was set up incorrectly. Maybe the viewing conditions were wrong, or there was some unnoticed interference with the experiment. In the case of the balloon, if it still sticks to the wall, there might still be truth in the theory that rubbing scrapes off old rubber and exposes a fresh, sticky surface, but in the testing process the balloon wasn't left long enough to dry, or the air was too cold and drying was retarded. These could be responsible for the expected result failing to appear, and until these issues are settled, we are not forced to reject the theory.

The form of the deduction in this revised hypothetico-deductive model now has three components, the hypothesis (that is, the theory being tested), the conditions of testing, and the predicted effect: If the hypothesis is true, then if conditions are right the effect will be observed. If in fact the effect is not observed, then we are forced to conclude that *either* the hypothesis is false *or* the conditions are not right. Blame the conditions if you want to retain the hypothesis. If there are too few neutrino-indicative clicks, maybe the counter or the argon-separation process is inefficient. The fluid may be contaminated or perhaps the tank is not deep enough or it is too deep in the earth. Lots of things can go wrong with experiments, and failed predictions do not automatically indicate a false theory.

The effect might be there, in other words, but we are just not seeing it because we are not doing the experiment properly. This is not an unreasonable excuse, given the complication and sensitivity of most scientific experiments. It's not so unreasonable in simple cases either. Theories of gravity predict that if two objects of unequal weight are dropped simultaneously from the same height, they will hit the ground at the same time. But if you try

the experiment it turns out that they don't hit at the same time. The problem is not with the theory. The problem is that the experimental condition (there must be no air resistance) is not satisfied.

As long as the experimental conditions are observable, however, they can be checked, and blaming conditions is not unresolvable speculation. It is rather a suggestion to have a look and see, and to do the experiment again with the conditions right. Drop the two things in a vacuum. Check the efficiency of the counters. That is the business of the experimentalist, to keep tabs on the conditions and to specify standards by which to tell when conditions are right. So there are limits to how far you can get by blaming the experimental conditions. This limitation, though, will be less effective on the next detail that must be added to the H-D model of testing.

If we think back on the example, it is clear that more than one theoretical claim is needed to deduce the predictions. The hypothesis alone does not entail the prediction; it is rather the hypothesis in conjunction with **auxiliary theories** from which predictions are deduced. The realistic model of testing begins with the deduction that if the hypothesis is true *and* other theories that describe mechanisms relevant to the manifestation of the effect are true, then if conditions are right the effect will be observed. Of all these theories used to deduce the prediction, there will be one theory distinguished as the hypothesis, that is, the theory we are interested in testing, but it is important to keep in mind that this is a distinction that depends on the interests of the scientists. All the theories used in the deduction, hypothesis and auxiliaries alike, are about unobservables, and in this sense all are taking risks and are at some point in need of confirmation.

Consider the case of solar neutrinos where the hypothesis is the claim that nuclear fusion is at work in the solar interior. This claim by itself does not entail any predictions about neutrinos or detector clicks. The hypothesis doesn't even mention neutrinos.

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The deduction of the crucial prediction can be done only with additional premises. There is a premise to the effect that fusion creates neutrinos, and this claim is a theory, one of the auxiliary theories needed for the deduction. Furthermore, they are not just any old neutrinos but specifically electron neutrinos (there are two other kinds) produced in the sun. Once produced, the neutrinos are sufficiently elusive to escape from the solar depths where they are created and to speed to earth. This is another theory required for the deduction. There must also be a theoretical account of how the neutrinos interact in the tank of cleaning fluid, how the argon is chemically separated, and so on. All of this theoretical information is needed to draw the experimental consequence from the solar theory. That's why science isn't so easy that you and I can settle in an instant the issue of nuclear fusion in the sun by simply looking for neutrinos in our cleaning fluid. You have to know a lot of theory in order to do meaningful experiments.

The logical structure of the deduction, then, is that if *all* of these theories are true, the theory of solar fusion and of fusion producing neutrinos and of the great mobility of neutrinos and so on, then the effect must happen. If all goes well and the predicted effect does happen, this contributes to the inductive evidence in favor of the hypothesis, the claim about the source of solar energy. This complicated case with auxiliaries and experimental conditions is not a new case. It is a more faithful representation of every case, an improvement on the original, basic model of H-D confirmation. A positive test is not unambiguous confirmation of the hypothesis, but it helps.

What if the effect doesn't happen? Suppose, as is in fact the case, there aren't nearly enough neutrinos observed and yet all the prescribed experimental conditions check out as being fulfilled. The deductive argument of falsification forces the conclusion that, without a doubt, not all the theories, hypothesis plus auxiliaries, are true. That much, but only that much, is known for

sure. It is unlikely that they are all false, and surely this extreme conclusion is not warranted by the evidence. But is it the hypothesis that is false or one of the auxiliaries? Don't forget that the auxiliaries, even if previously tested, have not been unambiguously confirmed. They are not above suspicion. The negative result of the experiment deductively proves that one or more of the theories is false, but it does not, indeed it cannot, indicate which one it is. There is no unambiguous falsification of the hypothesis. By blaming an auxiliary we can retain the hypothesis, and the possibility of its being true can be kept alive in the face of a false prediction.

This is exactly what is happening these days in the solar-neutrino problem. Virtually nobody wants to reject the theory about nuclear fusion as the energy source of stars. It is a firmly entrenched account that explains a lot about stars, the abundance of the elements in the universe, and the evolution of the universe. Rejecting this theory would force a great deal of rewriting in physics and astronomy. Some of the auxiliary theories, on the other hand, are not so securely entrenched, and their rejection would upset less in the web of scientific knowledge. For example, the claim that the neutrinos travel virtually uninhabited from the sun to the earth could be wrong. There could be some unseen matter, the speculated weakly interacting massive particles (yes, they are called WIMPS), interfering with the neutrinos before they can get here. Or the theoretical claim that the neutrinos escape unscathed from the center of the sun could be mistaken. Any subtle influence that changed them from electron neutrinos into either muon or tau neutrinos (and such changes are theoretically possible) would render the neutrinos invisible to the detectors presently used. These are the sorts of theoretical claims, the auxiliaries, that are drawing the doubt as a result of the solar-neutrino problem.

It is important to understand the difference between the auxiliary theories and the experimental conditions. Though either

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or both can absorb the blame for a falsified prediction, the conditions are directly checkable (and often correctable) whereas the auxiliaries are not. The inclusion of auxiliaries introduces a lingering ambiguity in cases of falsification, since the auxiliary theories are not directly checkable. The observations are never clear as to which of the theories, hypothesis or auxiliaries, is the culprit, and this reveals the importance of internal virtues. The decision about which of the theories to question first is often guided, as in the solar-neutrino case, by consideration of internal features such as entrenchment, explanatory coherence, and simplicity.

The realization that the testing of a theory against the observable evidence can yield neither certain confirmation nor unequivocal falsification indicates the impossibility of crucial experiments. A crucial experiment would be a test that takes place in the context of two rival theories that offer conflicting descriptions of the same aspect of the world. Light-as-a-wave or light-as-a-particle was once a question of competing theories. Earth-centered or sun-centered universe was another. And without stretching the idea too much, any theory (there are x 's) and its denial (there aren't x 's) are rival hypotheses. A crucial experiment would be a definitive test that would clearly rule out the possibility of one theory and thereby rule in the other. But such clarity never happens, and expecting it is the product of a naive view of the falsification process. Either hypothesis is retainable regardless of the experimental outcome. It may require rejecting or even adding some auxiliary claims, but it can be done.

This is not to say that we never decide or cannot decide between rival theories. It happens all the time. The point is simply that the observational evidence alone does not force the decision.

Here is a final but classic example of retaining a hypothesis in the face of a failed prediction. Not only were the auxiliaries doubted before the hypothesis, but a new, ad hoc auxiliary theory

was devised to account for the unfavorable experimental outcome. The whole affair had a very profitable result.

In the nineteenth century, Newton's theory of gravity was tested through its predictions of astronomical events. The theory (or, more carefully, the hypothesis, since it was the theory being tested) predicted eclipses, the comings and goings of comets, and the orbits of the planets. In fine H-D form, the theory was put to the test by observations of the predicted positions of the planets, all seven of them. The theory made lots of true predictions and accumulated lots of inductive, H-D support. But then it made a mistake. The predicted orbital positions of the planet Uranus did not match the observed positions. But this false prediction did not force the rejection of Newton's theory, as our preliminary, naive account of falsification would have suggested.

To retain the theory, though, one must fix the blame for the false prediction either on the experimental conditions, saying the prediction isn't really false (things just looked that way), or on the auxiliaries, saying the prediction is false but the hypothesis (Newton's theory) is not to blame. Appealing to the experimental conditions was not a lasting option, as the measurements and observations were demonstrably accurate enough to show that Uranus really wasn't where it should be.

The auxiliaries then. What are they? To calculate the position of a planet one must know not only about the gravitational force that affects it but also that there are no other forces of consequence acting on it. It is also necessary to know the sizes and positions of the nearby massive objects causing the gravitational forces. In this case that means a model of the solar system, and it was this model that drew suspicion. Even if the gravitational force works exactly as the theory says, we will get erroneous predictions if we do the calculations with erroneous data on the objects causing the gravity. Errors of this kind would result if the mass of Uranus is misunderstood, or if we are wrong about its orbital radius. Or maybe there is yet an eighth, as yet unseen, planet be-

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yond Uranus, which is pulling it from its predicted orbit. In fact, the position of this putative planet, where it would have to be to cause the deviation, was calculated, and in 1846 Neptune was discovered, right where it should be.

The decision to stand by Newton's theory of gravitation and to question the auxiliary theories instead was guided by internal factors. Newton's theory was richly explanatory and elegantly efficient. It made good sense of a lot of what goes on in the universe, and rejecting it would force a great deal of rethinking and reworking of the understanding of the physical world. The model of the solar system, on the other hand, was not so crucially connected to the theoretical understanding of the universe. It could be rather easily revised, and so it was. The decision to question this particular auxiliary theory rather than the hypothesis in turn influenced the selection of further tests; thus internal factors direct not only the justification of theories but the choice of experiments as well.

SUMMARY

The moral of the story is threefold.

Theories are compared to the evidence in groups, never one at a time. In the most realistic model of scientific testing, the hypothesis and the auxiliaries are equally on the line. They are equally vulnerable to the data. As a result of this, science admits of no unambiguous falsification just as it has no facility for unequivocal, dead-certain confirmation of theories. It is always the consequences of a theory that are compared to the evidence, and between theory and its consequences there is always room to equivocate.

The second important point in the analysis of confirmation has already slipped out. In science, there is no conclusive proof or disproof of a theory. Scientific proof is not conclusive proof. This is not to say that there is no justification of scientific theories.

Indeed, the likelihood of a theory being true is influenced by the positive or negative outcome of a test, as well as by the evaluation of its internal virtues. But the loss of certainty should be a warning to the superscrupulous who expect nothing less than certainty from science. If we hold out for certainty, we get nothing, and we will miss the crucial activity of science that is the evaluation of degrees of likelihood of theories and the warrant for believing that they are true. It is a comparative analysis, assessing not absolute justification of theories but a relative scale by which to judge among competing claims.

Finally, it has become clear that science must rely on more than just the observable evidence to adjudicate the warrant for theories. Science is much more than simply letting the facts decide. Internal factors are an essential and influential part of the justification process. They must be, since the observable data do not contain enough information to determine which theories are true. The observable evidence, as it is sometimes said, underdetermines the theoretical account of the world.